

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date:

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name:

Reg. no.:

1. Use MongoDB to implement the following DB operations

1. Create a database called 'vehicles' and write a MongoDB query to select database as "vehicles".

```
animals> use vehicle
switched to db vehicle
vehicle>
```

2. Write a MongoDB query to display all the databases

```
animals> show dbs
admin          40.00 KiB
animal         80.00 KiB
backend_db     40.00 KiB
config         100.00 KiB
local          0.00 KiB
local          120.00 KiB
my_database    80.00 KiB
otp_db         100.00 KiB
vehicles       100.00 KiB
animals>
```

3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.

```
vehicles> // Capped collection (size in bytes)
... db.createCollection("two_wheelers", { capped: true, size: 5242880, max: 5000 });
...
... // Standard collection
... db.createCollection("four_wheelers");
```

4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```
vehicles> db.two_wheelers.insertMany([
... { bike_name: "Splendor Plus", model: "gear", category: "100cc", colors_available: ["black", "purple"], manufacturer: "Hero", performance:
7, timestamp: new Date("2021-05-18"), price: 75000 },
... { bike_name: "Activa 6G", model: "gearless", category: "125cc", colors_available: ["blue", "red", "white"], manufacturer: "Honda", performanc
e: 8, timestamp: new Date("2021-01-15"), price: 85000 },
... { bike_name: "Pulsar NS200", model: "gear", category: "200cc", colors_available: ["black", "sport red"], manufacturer: "Bajaj", performanc
e: 9, timestamp: new Date("2021-03-20"), price: 158000 },
... { bike_name: "FZ-S", model: "gear", category: "150cc", colors_available: ["matte blue", "black"], manufacturer: "Yamaha", performance: 8,
timestamp: new Date("2021-11-12"), price: 120000 },
... { bike_name: "Jupiter", model: "gearless", category: "110cc", colors_available: ["brown", "grey"], manufacturer: "TVS", performance: 7, ti
imestamp: new Date("2022-06-05"), price: 80000 }
... ]);
```

5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel

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etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```
vehicles> db.four_wheelers.insertMany([
...   { vehicle_name: "Swift", model: "own", category: "car", variants: ["vxi", "zxi", "petrol"],
timestamp: new Date("2022-04-10"), price: 700000 },
...   { vehicle_name: "Eicher Pro", model: "commercial", category: "heavy truck", variants: ["diesel"],
timestamp: new Date("2020-08-22"), price: 2500000 },
...   { vehicle_name: "Fortuner", model: "own", category: "car", variants: ["diesel", "4x4"],
timestamp: new Date("2023-01-05"), price: 3500000 },
...   { vehicle_name: "Ace Gold", model: "commercial", category: "mini truck", variants: ["diesel"],
timestamp: new Date("2021-09-15"), price: 500000 },
...   { vehicle_name: "City", model: "own", category: "car", variants: ["vxi", "petrol"],
timestamp: new Date("2022-12-10"), price: 1200000 }
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699fcf5b9c35c57c29cebeb3'),
    '1': ObjectId('699fcf5b9c35c57c29cebeb4'),
    '2': ObjectId('699fcf5b9c35c57c29cebeb5'),
    '3': ObjectId('699fcf5b9c35c57c29cebeb6'),
    '4': ObjectId('699fcf5b9c35c57c29cebeb7')
  }
}
```

6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.

```
{
  vehicle_name: 'Ashok Leyland Best',
  model: 'commercial',
  category: 'mini truck',
  variants: [ 'diesel' ],
  manufacturer: 'Ashok Leyland',
  performance: 7,
  timestamp: ISODate('2023-04-01T00:00:00.000Z'),
  price: 900000
},
{
  _id: ObjectId('699fcs6553494fc219cebead'),
  vehicle_name: 'Hyundai i20',
  model: 'own',
  category: 'car',
  variants: [ 'petrol', 'diesel' ],
  manufacturer: 'Hyundai',
  performance: 9,
  timestamp: ISODate('2023-06-25T00:00:00.000Z'),
  price: 850000
},
{
  _id: ObjectId('699fcf5b9c35c57c29cebeb3'),
  vehicle_name: 'Swift',
  model: 'own',
  category: 'car',
  variants: [ 'vxi', 'zxi', 'petrol' ],
  manufacturer: 'Maruti',
  performance: 8,
  timestamp: ISODate('2022-04-10T00:00:00.000Z'),
  price: 700000
},
{
  _id: ObjectId('699fcf5b9c35c57c29cebeb0'),
  vehicle_name: 'Eicher Pro',
  model: 'commercial',
  category: 'heavy truck',
  variants: [ 'diesel' ],
  manufacturer: 'Eicher',
  performance: 6,
  timestamp: ISODate('2020-08-22T00:00:00.000Z'),
  price: 2500000
}
```

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```
timestamp: ISODate('2020-08-22T00:00:00.000Z'),
price: 2500000
},
{
  _id: ObjectId('6094cf5b9c35c57c29eebb5'),
  vehicle_name: 'Fortuner',
  model: 'van',
  category: 'car',
  variants: [ 'diesel', '4x4' ],
  manufacturer: 'Toyota',
  performance: 9,
  timestamp: ISODate('2023-01-05T00:00:00.000Z'),
  price: 3500000
},
{
  _id: ObjectId('6094cf5b9c35c57c29eebb6'),
  vehicle_name: 'Axe Gold',
  model: 'commercial',
  category: 'mini truck',
  variants: [ 'petrol', 'cng' ],
  manufacturer: 'Tata',
  performance: 6,
  timestamp: ISODate('2021-09-15T00:00:00.000Z'),
  price: 500000
},
{
  _id: ObjectId('6094cf5b9c35c57c29eebb7'),
  vehicle_name: 'City',
  model: 'van',
  category: 'car',
  variants: [ 'vci', 'petrol' ],
  manufacturer: 'Honda',
  performance: 8,
  timestamp: ISODate('2022-12-10T00:00:00.000Z'),
  price: 1200000
}
}
vehicles>
```

- Write a MongoDB query to display only vehicle name and price in all the collection of the database

```
vehicles> db.two_wheelers.find({}, { bike_name: 1, price: 1, _id: 0 });
... db.four_wheelers.find({}, { vehicle_name: 1, price: 1, _id: 0 });
{ vehicle_name: 'Maruti Swift', price: 700000 },
{ vehicle_name: 'Tata Intra', price: 800000 },
{ vehicle_name: 'Mahindra Bolero', price: 950000 },
{ vehicle_name: 'Ashok Leyland Gust', price: 900000 },
{ vehicle_name: 'Hyundai i20', price: 850000 },
{ vehicle_name: 'Swift', price: 700000 },
{ vehicle_name: 'Eicher Pro', price: 2500000 },
{ vehicle_name: 'Fortuner', price: 3500000 },
{ vehicle_name: 'Axe Gold', price: 500000 },
{ vehicle_name: 'City', price: 1200000 }
}
vehicles>
```

- Write a MongoDB query to display two_wheelers from a particular company

```
vehicles> db.two_wheelers.find({ manufacturer: 'Honda' });
{
  _id: ObjectId('6094cf5b9c35c57c29eebb5'),
  bike_name: 'Honda Activa 6G',
  model: 'gearless',
  category: '125cc',
  colors_available: [ 'black', 'white', 'blue' ],
  manufacturer: 'Honda',
  performance: 7,
  timestamp: ISODate('2023-06-10T00:00:00.000Z'),
  price: 65000
},
{
  _id: ObjectId('6094cf5b9c35c57c29eebb6'),
  bike_name: 'Activa 6G',
  model: 'gearless',
  category: '125cc',
  colors_available: [ 'blue', 'red', 'white' ],
  manufacturer: 'Honda',
  performance: 8,
  timestamp: ISODate('2022-01-10T00:00:00.000Z'),
  price: 60000
}
}
vehicles>
```

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9. Write a MongoDB query to display four_wheelers available in diesel variants

```
vehicles> db.four_wheelers.find({ variants: "diesel" });
{
  "_id": ObjectId("699fc66553494fc225c6b5aa"),
  "vehicle_name": "Tata Intra",
  "model": "commercial",
  "category": "mini truck",
  "variants": [ "diesel" ],
  "manufacturer": "Tata",
  "performance": 7,
  "timestamp": ISODate("2021-05-12T00:00:00.000Z"),
  "price": 800000
},
{
  "_id": ObjectId("699fc66553494fc225c6b5ab"),
  "vehicle_name": "Mahindra Bolero",
  "model": "van",
  "category": "car",
  "variants": [ "diesel" ],
  "manufacturer": "Mahindra",
  "performance": 6,
  "timestamp": ISODate("2020-07-18T00:00:00.000Z"),
  "price": 950000
},
{
  "_id": ObjectId("699fc66553494fc225c6b5ac"),
  "vehicle_name": "Ashok Leyland Dust",
  "model": "commercial",
  "category": "mini truck",
  "variants": [ "diesel" ],
  "manufacturer": "Ashok Leyland",
  "performance": 7,
  "timestamp": ISODate("2023-04-11T00:00:00.000Z"),
  "price": 900000
},
{
  "_id": ObjectId("699fc66553494fc225c6b5ad"),
  "vehicle_name": "Ashok Leyland",
  "model": "commercial",
  "category": "mini truck",
  "variants": [ "diesel" ],
  "manufacturer": "Ashok Leyland",
  "performance": 7,
  "timestamp": ISODate("2023-04-11T00:00:00.000Z"),
  "price": 900000
}
```

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

```
vehicles> // For two-wheelers
... db.two_wheelers.find({ performance: { $gt: 5 } }, { bike_name: 1, category: 1, manufacturer: 1, _id: 0 });
...
// For Four-wheelers
... db.four_wheelers.find({ performance: { $gt: 5 } }, { vehicle_name: 1, category: 1, manufacturer: 1, _id: 0 });
{
  "vehicle_name": "Maruti Swift",
  "category": "car",
  "manufacturer": "Maruti"
},
{
  "vehicle_name": "Tata Intra",
  "category": "mini truck",
  "manufacturer": "Tata"
},
{
  "vehicle_name": "Mahindra Bolero",
  "category": "car",
  "manufacturer": "Mahindra"
},
{
  "vehicle_name": "Ashok Leyland Dust",
  "category": "mini truck",
  "manufacturer": "Ashok Leyland"
},
{
  "vehicle_name": "Hyundai i20",
  "category": "car",
  "manufacturer": "Hyundai"
},
{
  "vehicle_name": "Swift", "category": "car", "manufacturer": "Maruti" },
{
  "vehicle_name": "Eicher Pro",
  "category": "mini truck",
  "manufacturer": "Eicher"
}
```

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2. Use MongoDB to implement the following DB operations for a Zoo

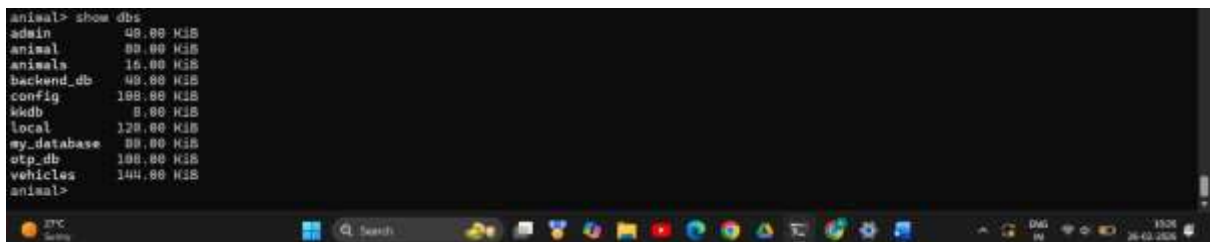
1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.

```
vehicles> use animal
switched to db animal
animal>
```



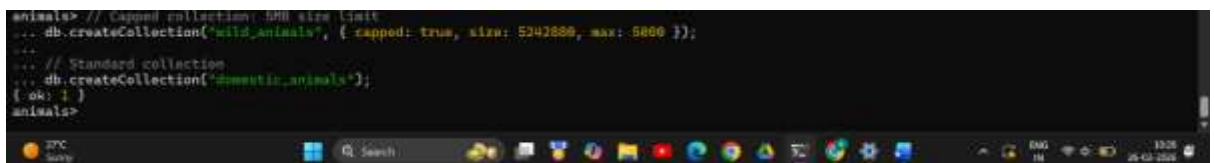
2. Write a MongoDB query to display all the databases.

```
animal> show dbs
admin          48.00 KIB
animal         80.00 KIB
animals        16.00 KIB
backend_db     48.00 KIB
config         108.00 KIB
local          8.00 KIB
local          128.00 KIB
my_database    88.00 KIB
otp_db         108.00 KIB
vehicles       144.00 KIB
animal>
```



3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.

```
animals> // Capped collection: 5MB size limit
... db.createCollection("wild_animals", { capped: true, size: 5242880, max: 5000 });
...
... // Standard collection
... db.createCollection("domestic_animals");
{ok: 1}
animals>
```



4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless),

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favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the anim

```
animals> // Capped collection: 5MB size limit
... db.createCollection('wild_animals', { capped: true, size: 5242880, max: 5000 });
... // Standard collection
... db.createCollection('domestic_animals');
( out: 1 )
animals> db.wild_animals.insertMany([
... { animal_name: "Lion", nature: "harm", favorite_foods: ["meat", "deer"], care_taker_name: "John Doe", life_span: 10, timestamp: new Date(),
... expenses: 5000 },
... { animal_name: "Elephant", nature: "harmless", favorite_foods: ["sugar cane", "grass"], care_taker_name: "Alice Smith", life_span: 60, timestamp: new Date(),
... expenses: 8000 },
... { animal_name: "Tiger", nature: "harm", favorite_foods: ["meat", "wild boar"], care_taker_name: "John Doe", life_span: 12, timestamp: new Date(),
... expenses: 3500 },
... { animal_name: "Wolf", nature: "harm", favorite_foods: ["rabbits", "meat"], care_taker_name: "Ben Wilson", life_span: 5, timestamp: new Date(),
... expenses: 2000 },
... { animal_name: "Hedgehog", nature: "harmless", favorite_foods: ["leaves", "insects"], care_taker_name: "Alice Smith", life_span: 25, timestamp: new Date(),
... expenses: 6000 }
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('609f42af3c35c67e29cebe07'),
    '1': ObjectId('609f42af3c35c67e29cebe08'),
    '2': ObjectId('609f42af3c35c67e29cebe09'),
    '3': ObjectId('609f42af3c35c67e29cebe0a'),
    '4': ObjectId('609f42af3c35c67e29cebe0b')
  }
}
animals>
```

al registered at the Zoo) and expenses.

5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

```
animals> db.domestic_animals.insertMany([
... { animal_name: "Rabbit", gender: "female", favorite_foods: ["carrots", "grass"], animal_petname: "Bunny", life_span: 9, timestamp: new Date(),
... expenses: 500 },
... { animal_name: "Goat", gender: "male", favorite_foods: ["grass", "leaves"], animal_petname: "Billy", life_span: 12, timestamp: new Date(),
... expenses: 1000 },
... { animal_name: "Sheep", gender: "female", favorite_foods: ["grass"], animal_petname: "Shaun", life_span: 4, timestamp: new Date(), expense
... expenses: 800 },
... { animal_name: "Cow", gender: "female", favorite_foods: ["hay", "hush"], animal_petname: "Daisy", life_span: 20, timestamp: new Date(), ex
... expenses: 2000 },
... { animal_name: "Dog", gender: "male", favorite_foods: ["kibble", "meat"], animal_petname: "Rucky", life_span: 10, timestamp: new Date(), e
... expenses: 1200 }
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('609f438c3c35c67e29cebe0d'),
    '1': ObjectId('609f438c3c35c67e29cebe0e'),
    '2': ObjectId('609f438c3c35c67e29cebe0f'),
    '3': ObjectId('609f438c3c35c67e29cebe10'),
    '4': ObjectId('609f438c3c35c67e29cebe11')
  }
}
animals>
```

6. Write a MongoDB query to display all documents available in wild_animals and domestic_animals.

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```
animals> db.wild_animals.find().pretty();
... db.domestic_animals.find().pretty();
{
  "_id": ObjectId("699fd38c9c35c57c29cebebd"),
  "animal_name": "Rabbit",
  "gender": "female",
  "favorite_foods": [ "carrots", "grass" ],
  "animal_petname": "Bunny",
  "life_span": 9,
  "timestamp": ISODate("2026-02-26T04:58:52.656Z"),
  "expenses": 500
},
{
  "_id": ObjectId("699fd38c9c35c57c29cebebe"),
  "animal_name": "Goat",
  "gender": "male",
  "favorite_foods": [ "grass", "leaves" ],
  "animal_petname": "Billy",
  "life_span": 12,
  "timestamp": ISODate("2026-02-26T04:58:52.656Z"),
  "expenses": 1000
},
{
  "_id": ObjectId("699fd38c9c35c57c29cebeb9"),
  "animal_name": "Sheep",
  "gender": "female",
  "favorite_foods": [ "grass" ],
  "animal_petname": "Shaun",
  "life_span": 4,
  "timestamp": ISODate("2026-02-26T04:58:52.656Z"),
  "expenses": 800
},
{
  "_id": ObjectId("699fd38c9c35c57c29cebec0"),
```

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database

```
animals> db.wild_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 });
... db.domestic_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 });
{ animal_name: 'Rabbit', expenses: 500 },
{ animal_name: 'Goat', expenses: 1000 },
{ animal_name: 'Sheep', expenses: 800 },
{ animal_name: 'Cow', expenses: 2000 },
{ animal_name: 'Dog', expenses: 1200 }
```

8. Write a MongoDB query to display domestic_animals whose life is a particular year

```
animals> db.domestic_animals.find({ life_span: 12 });
{
  "_id": ObjectId("699fd38c9c35c57c29cebebe"),
  "animal_name": "Goat",
  "gender": "male",
  "favorite_foods": [ "grass", "leaves" ],
  "animal_petname": "Billy",
  "life_span": 12,
  "timestamp": ISODate("2026-02-26T04:58:52.656Z"),
  "expenses": 1000
}
```

9. Write a MongoDB query to display wild_animals available under a particular care_taker

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```
animals> db.wild_animals.find({ care_taker_name: "John Doe" });
[
  {
    _id: ObjectId('608fd2af9c35c57c29ceb6b6'),
    animal_name: 'Lion',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer' ],
    care_taker_name: 'John Doe',
    life_span: 15,
    timestamp: ISODate('2026-02-26T04:57:19.483Z'),
    expenses: 5000
  },
  {
    _id: ObjectId('608fd2af9c35c57c29ceb6ba'),
    animal_name: 'Tiger',
    nature: 'harm',
    favorite_foods: [ 'meat', 'wild boar' ],
    care_taker_name: 'John Doe',
    life_span: 12,
    timestamp: ISODate('2026-02-26T04:57:19.483Z'),
    expenses: 5000
  }
]
animals>
```

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

```
animals> // For Wild Animals
... db.wild_animals.find({ life_span: { $gt: 5 } }, { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 });
...
// For Domestic Animals
... db.domestic_animals.find({ life_span: { $gt: 5 } }, { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 });
[
  {
    animal_name: 'Rabbit',
    favorite_foods: [ 'carrots', 'grass' ],
    expenses: 500
  },
  {
    animal_name: 'Goat',
    favorite_foods: [ 'grass', 'leaves' ],
    expenses: 1000
  },
  {
    animal_name: 'Cow',
    favorite_foods: [ 'hay', 'husk' ],
    expenses: 2000
  },
  {
    animal_name: 'Dog',
    favorite_foods: [ 'kibble', 'meat' ],
    expenses: 1200
  }
]
animals>
```